

ICT ACADEMY – CYBER SECURITY PROGRAM

Day 1 – Networking Fundamentals & AWS EC2 Practical Lab

Learning Log / GitHub Documentation

Session Details

Participant	Mohammad Jawad
Day	Day 1
Date	25 September 2026
Hands-on Platform	Amazon Web Services (AWS)
Cloud Service Used	Amazon EC2
Operating System	Amazon Linux 2023

Day 1 Focus

Build the basic networking foundation and apply it through a real cloud lab by launching and connecting to an EC2 Linux server. This document converts the handwritten Day 1 notes and practical experience into a clean, beginner-friendly learning record suitable for GitHub.

1. Day 1 Learning Overview

Area	What was learned
Networking	Network basics, OSI model, IP addresses, subnetting, routing and routing protocols.
IP Addressing	IPv4, public vs. private IP addresses, classful IP ranges and the purpose of a subnet mask.
Ports	What a port number is and how it identifies the application/service receiving network traffic.
Servers	Basic types of servers such as web, application, database, email, game, proxy and reverse proxy servers.
Cybersecurity	Security domains, CIA triad, physical/digital data protection and ethical hacking team roles.
Cloud Practical	Created an AWS account, launched an EC2 instance, connected through SSH and worked with Amazon Linux 2023.

2. Networking Fundamentals

Network: A group of computers or electronic devices connected so they can communicate and share resources.

OSI Model – 7 Layers

Layer	Main purpose
7. Application	Provides network services to user applications.
6. Presentation	Data format, translation, encryption and compression.
5. Session	Establishes, manages and terminates communication sessions.
4. Transport	End-to-end delivery, reliability and flow control.
3. Network	Logical addressing and routing between networks.
2. Data Link	Frames, MAC addressing and local-link delivery.
1. Physical	Transmission of bits through physical media/signals.

IP Addressing

- **IP address:** A logical address used to identify a device/interface on an IP network.
- **IPv4:** Uses 32 bits and is normally written as four decimal octets, for example 192.168.1.10.
- **Private IP:** Used inside private networks such as home, campus and enterprise LANs. Common private IPv4 blocks include 10.0.0.0/8, 172.16.0.0/12 and 192.168.0.0/16.
- **Public IP:** Globally routable address used for communication across the public Internet. A public address may be assigned by an Internet/service provider.
- **Subnet mask:** Helps identify which part of an IP address represents the network and which part represents hosts.

Classful IPv4 Ranges – Historical Classification

Class	First-octet range	Traditional purpose
A	1–126	Very large networks
B	128–191	Medium-sized networks
C	192–223	Smaller networks
D	224–239	Multicast
E	240–255	Experimental / reserved

Note

Modern networks primarily use CIDR/subnetting rather than the old classful A/B/C allocation model.

3. Routing, Protocols and Ports

Routing: The process of selecting and forwarding packets from a source network toward a destination network.

Routing protocol: A set of rules used by routers to exchange information and determine routes.

Routing Protocols

Protocol	Full form / role
RIP v1	Routing Information Protocol; a distance-vector routing protocol.
RIP v2	Updated RIP version with improvements such as support for subnet information.
EIGRP	Enhanced Interior Gateway Routing Protocol; an advanced routing protocol associated with Cisco networks.
IGRP	Interior Gateway Routing Protocol; an older Cisco routing protocol.
BGP	Border Gateway Protocol; used to exchange routing information between autonomous systems on the Internet.

Ports

A port number is a logical number used by TCP/UDP to identify the application or service that should receive network traffic. A useful analogy: **IP address = building address; port number = room number.**

Servers Discussed

Web server • Web application server • Database server • Email server • Gaming server • Load balancer • Proxy server • Reverse proxy server • Network server • File server

4. Cybersecurity Fundamentals

Cybersecurity is the practice of protecting systems, networks, applications and data from unauthorized access, misuse, disruption or damage.

CIA Triad

Confidentiality	Only authorized people can access information.
Integrity	Information remains accurate and unaltered.
Availability	Authorized users can access systems/data when needed.

Ethical Hacking Team Roles

Team	General role
Red Team	Offensive security: simulates attacks to identify weaknesses.
Blue Team	Defensive security: detects, responds to and protects against attacks.
Purple Team	Collaboration between offensive and defensive teams to improve security.

Security Areas Introduced

Physical security • Network security • Application security • IAM (Identity and Access Management) • Cloud security • Data security • Email security • OS security • Container security • Embedded security • Firmware security • Hardware security

OT/ICS security: Protection of Operational Technology and Industrial Control Systems such as PLCs and SCADA used in industrial environments.

5. Data, Operating Systems and Web Basics

Physical data/storage examples: NAS (Network Attached Storage), SAN (Storage Area Network), file servers, pen drives and hard disks.

Digital/cloud storage examples: SaaS-based storage, OneDrive, Google Drive, virtual file services and Amazon S3 in AWS.

GRC (Governance, Risk and Compliance): A security discipline concerned with managing security responsibilities, risks and compliance requirements.

Operating Systems: An OS manages computer hardware and provides services for applications. The notes distinguished client/end-user operating systems (e.g., Windows) from server operating systems (e.g., Linux-based) used for network/server workloads.

Web Request Flow – Basic Idea

Step	Action
1	User enters a URL in a browser.
2	DNS helps resolve the domain name to an IP address.
3	The client establishes network communication with the destination server.
4	The web server/application processes the request.
5	The server sends a response.
6	The browser renders the returned content.

6. Practical Lab – AWS EC2

Goal: Create a cloud-based Linux server, connect to it securely through SSH and perform a basic server-side task.

Lab Steps

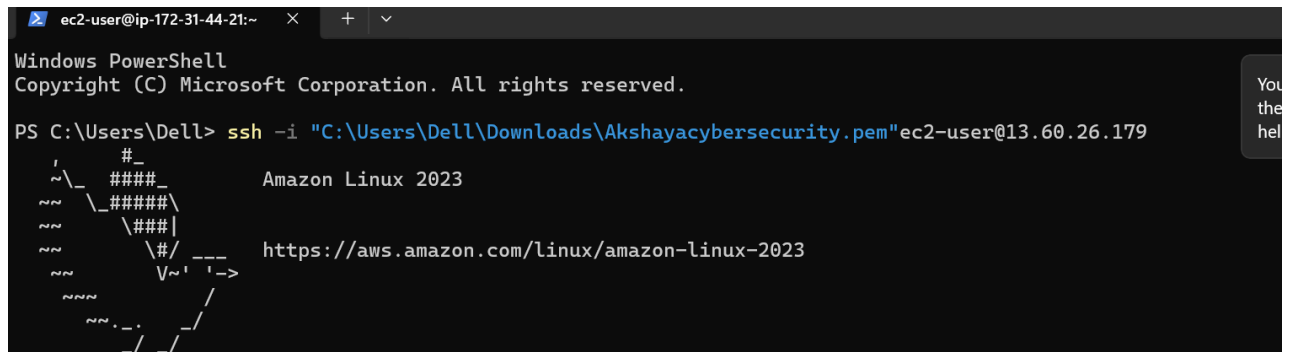
Step	Description
Step 1 – Prepare account	Sign in to AWS and open the AWS Management Console.
Step 2 – Open EC2	From AWS services, open EC2 (Elastic Compute Cloud). EC2 provides virtual machines called instances.
Step 3 – Launch an instance	Create a new EC2 instance and select Amazon Linux 2023 as the operating system image.
Step 4 – Configure the instance	Choose the required instance settings and create/select a key pair for secure SSH authentication.
Step 5 – Network access	Allow the required SSH access in the instance security group. Only open ports that are actually required.
Step 6 – Launch the server	Launch the instance and note its public IP address or public DNS name.
Step 7 – Connect using SSH	From Windows PowerShell, use the downloaded private key (.pem) to connect to the EC2 instance.
Step 8 – Verify Linux shell	After successful login, the terminal shows the Amazon Linux environment and an ec2-user shell prompt.
Step 9 – Practical task	Continue the lab by creating/serving the required content. The Day 1 practical ended with a bird image being displayed successfully as the output.

SSH Command Used in the Practical

```
ssh -i "path-to-your-key.pem" ec2-user@YOUR_EC2_PUBLIC_IP
```

Important: Never upload the .pem private key, passwords, access keys or other secrets to GitHub.

Practical Evidence – Successful EC2 SSH Login



The screenshot shows a successful SSH connection to an Amazon Linux 2023 EC2 instance, including the Linux shell prompt.

7. What I Learned on Day 1

1. Understood the basic purpose of computer networks and the OSI model.
2. Learned the difference between public and private IPv4 addresses.
3. Reviewed subnet masks and the basic idea of subnetting.
4. Understood what routing and routing protocols do.
5. Learned what port numbers are and why applications use them.
6. Identified common server types and their purposes.
7. Was introduced to cybersecurity domains and the CIA triad.
8. Understood the basic roles of Red Team, Blue Team and Purple Team.
9. Gained hands-on experience creating an AWS EC2 instance.
10. Connected to a cloud Linux server using SSH.
11. Practiced working with Amazon Linux 2023 in a real cloud environment.

8. Day 1 Practical Architecture

Laptop / Windows PowerShell -> Internet -> AWS Cloud -> EC2 Instance (Amazon Linux 2023)

9. GitHub Posting Structure

```
ICT-Academy-Cybersecurity-Journey/  
  Day-01-Networking-AWS-EC2/  
    Day-01-Report.pdf  
    screenshots/  
    README.md  
  Day-02-.../  
    README.md
```

Suggested GitHub Post Text

Day 1/13 – ICT Academy Cyber Security Program

Started my 13-day cybersecurity learning journey with networking fundamentals and a hands-on AWS EC2 lab. Today I learned about the OSI model, IPv4 addressing, subnetting, routing, ports, servers, cybersecurity basics and ethical hacking team roles. For the practical session, I created an AWS EC2 instance using Amazon Linux 2023, connected to it through SSH and completed a basic server-side task that produced a bird image as the final output.

Hands-on: AWS • EC2 • Amazon Linux 2023 • SSH • Networking

10. Next Step – Day 2

Continue the same format for each day: Concepts learned, Practical lab, Commands/tools used, Output/evidence, What I learned, Key takeaway. This will make the final 13-day GitHub repository easy for another beginner, recruiter or interviewer to understand.